

Step 1 — Embedding Integrity Check

Sanity diagnostic before any downstream evaluation. Catches degenerate output (zero variance, NaN, mismatched shape).

FM	n cells	dim	% NaN	cell-pair cos (mean / std)	Status
STATE	47977	2058	0.0	0.612 / 0.041	PASS
UCE	47977	1280	0.0	0.587 / 0.038	PASS
TranscriptFormer	47109	2048	0.0	0.643 / 0.042	PASS
Geneformer	46949	1152	0.0	0.621 / 0.039	PASS
scGPT	47977	512	0.0	0.598 / 0.044	PASS

NOTE: this diagnostic later caught a TF-Metazoa tokenization mismatch in the 6-FM extension (cell-pair cosine = 0.985, per-cell magnitude std = 0.001 → degenerate embedding).